Discover Chemistry

Review

A comprehensive review on the synthesis, properties, environmental impacts, and chemiluminescence applications of polystyrene (PS)

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Abstract
Polystyrene (PS), a widely used synthetic aromatic polymer, is integral to industries such as packaging, construction, and medical equipment. Its high molecular weight and economic advantages make it versatile, yet limitations like low oxygen barrier properties, vulnerability to smog, and a low melting point pose challenges. General Purpose PS (GPPS), Expanded PS (EPS), High Impact PS, (HIPS), and Syndiotactic PS (SPS) are some of the industrial important variants. This Study explores the synthesis, properties, production processes, and diverse applications of PS while addressing its disposal challenges and opportunities for improved reusability. Innovative approaches have transformed Waste PS (WPS) into valuable materials like ion exchangers, lightweight concrete, paints, and composites, enhancing environmental and economic value chains. A notable breakthrough is the use of WPS in chemiluminescent systems, where its photonic properties improve reaction sensitivity and diagnostic efficiency, achieving dual benefits of waste reduction and scientific advancement. Recycling WPS offers significant environmental benefits, reducing reliance on harmful disposal methods such as landfilling and incineration, thereby lowering pollution and greenhouse gas emissions. Advanced recycling technologies, including microwave-assisted pyrolysis and reverse polymerization, have improved efficiency and product recovery, unlocking the economic potential of WPS recycling. This aligns with Sustainable Development Goals 12 (Responsible Consumption and Production) and 13 (Climate Action). In a nutshell, the study encourages collaboration among researchers, policymakers, and industry stakeholders to establish robust policies, incentivize proper PS disposal and recycling, and promote innovations that enhance sustainability while mitigating environmental impacts.

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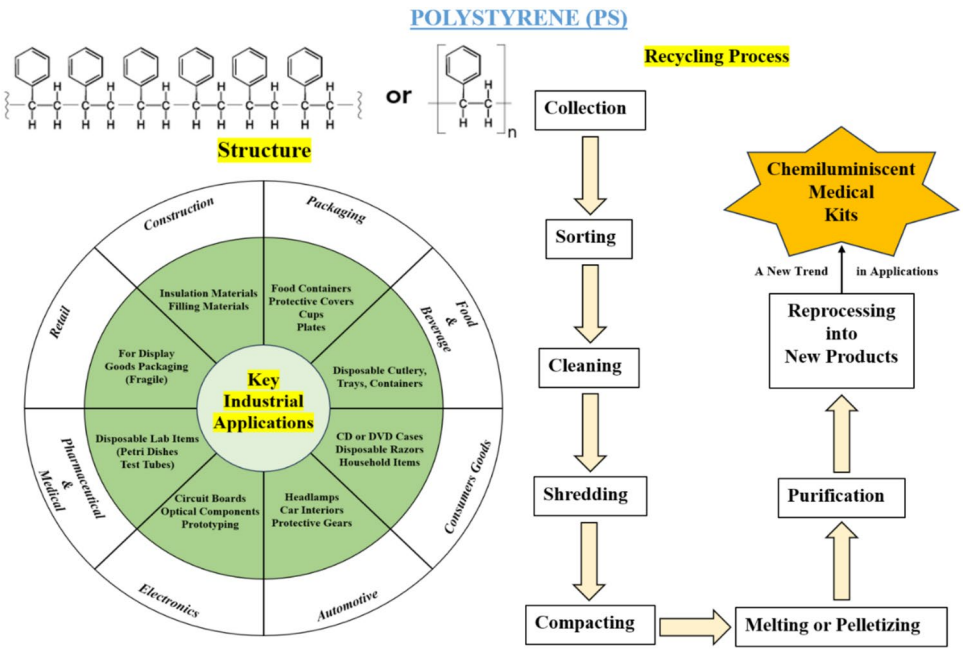
Graphical Abstract



Keywords Chemiluminescence · Environmental toxicity · Polystyrene · Sustainability · Synthesis methods

1 Introduction

Plastic consumption has increased significantly in developed nations over the last twenty to thirty years. These countries utilize scarce oil resources to manufacture a diverse array of plastics for numerous applications [1]. However, many of these products have a short life span of only a few months, leading to their disposal via open burning or in unsanitary landfills, which presents serious risks to both public health and the environment [2, 3]. The increased focus of industry is on reprocessing and recycling plastics due to environmental reasons, with several corporations innovating skills and strategies for plastic reutilizing [4]. Additionally, there is a growing effort to create biodegradable materials from plant sources as alternatives to conventional plastics. Since plastics are derived from unrenewable resources and are either non-decaying or decay at a slower rate, plastic waste often becomes a major form of persistent litter [5, 6], significantly shortening the lifespan of waste disposal facilities like landfills.

Per capita, plastic consumption in developing nations is significantly lower than in developed nations, yet the growth rate in plastic use is rapid [7, 8]. A lot of these plastics are made from expensive imported raw materials, which puts significant pressure on the country's foreign currency reserves. However, these regions have substantial potential for recycling and waste-to-product conversion for several reasons. Lower labor costs make operating recycling plants more affordable [9]. The presence of an informal sector remains suitable for limited reusing efforts, providing deprived urban populations with income opportunities. Regulatory standards for recycled materials [10] are often less stringent, and transportation costs are reduced due to primitive or inexpensive haulage methods. The lower cost of feedstocks can offer a competing advantage in production, and the pioneering use of scrap machinery can lower entry barriers for processing [11, 12]. In these developing nations, the options for plastic recycling are expanding as plastic consumption continues to grow. In the context of India, plastic waste management presents a significant environmental challenge [13]. Similar to global trends, India has experienced a dramatic increase in plastic production and consumption [14, 15]. As per India's Central Pollution Control Board (CPCB), the country creates nearly 3.5 million tons of plastic waste per annum [16–18], with the per capita plastic waste generation steadily rising. Of this waste, only about 60% is recycled, while the remaining 40% ends up in landfills, open dumps, or environmental litter [19, 20]. India's plastic waste management challenges

are exacerbated by rapid urbanization, population growth, and inadequate infrastructure [21, 22] for waste collection and processing. The informal recycling sector includes numerous small-scale recyclers and ragpickers who collect and segregate plastic waste, often without the proper facilities [23–25] or technologies for efficient and environmentally friendly recycling. In total, while India has made some progress in plastic waste management, significant challenges remain in terms of expanding recycling capacity, improving waste collection systems, and enhancing public awareness about plastic waste reduction and recycling.

Polystyrene (PS) is a synthetic aromatic polymer characterized by a high molecular weight, with the chemical formula $(C_8H_8)_n$, derived from the styrene monomer [26–30] (Fig. 1). The polymer can exist in rigid or foamed forms, whereas its monomer styrene remains liquid. General Purpose PS (GPPS) is known for being transparent, hard, and brittle [31, 32]. It is a cost-effective resin by weight, though it is not very effective as a barrier against oxygen and water vapor and has a comparatively low melting point. Code '6' identifies PS, according to the Society of Plastics Industry (SPI), a plastic commonly used in packaging [33–36]. Due to its low density, it tends to be easily blown away by the wind, causing significant contamination in the environment. Other dominant forms of PS utilized in the industry include Expanded PS (EPS), High-Impact PS (HIPS), Atactic PS (APS), Syndiotactic PS (SPS), and Isotactic PS (IPS) [37–39].

In the year 2023, India faces constant challenges with plastic waste management, generating about 4 million tons of plastic waste, reflecting an upward trend due to increased urbanization and economic growth [40–42]. Despite improvements, plastic waste being recycled primarily driven by the informal sector, still winds up in landfills, open dumps, or is poorly managed [43–45]. Multi-layered plastics (MLPs) and single-use plastics continue to pose substantial recycling challenges, despite governmental bans and the Plastic Waste Management Amendment Rules, 2022, which underline extended producer responsibility (EPR) [46–48]. Implementation of these regulations remains inconsistent, and compliance differs across states. Innovations such as advanced recycling technologies, including chemical recycling and AI-based waste segregation, are on the rise, along with public–private partnerships and investments in waste management infrastructure [49–52]. Public awareness campaigns promoting reduced plastic use and source segregation have gained traction, but wider behavioral change is slow, necessitating sustained efforts and education. While India has made progress in addressing its plastic waste issues in 2023, there remain significant gaps in achieving a completely sustainable and effective plastic waste management system [53–55], which requires continued collaborative efforts from the government, industry, and citizens. Thus, the main objectives of this study were; (1) to provide an outline of PS, its synthesis, properties, production, and current applications, (2) to identify disposal practices, and reusability of PS, (3) to address the studies related to PS in chemiluminescence, highlight research gaps, and propose future research directions.

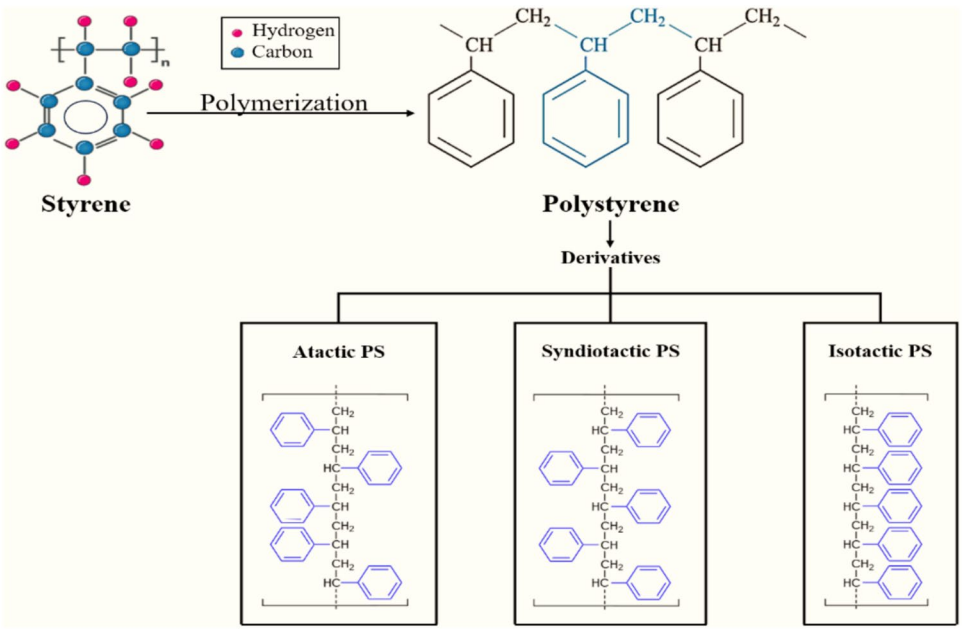


Fig. 1 Chemical structure of PS with its commercially important stereoisomer derivatives

1.1 Alignment of PS waste with SDG 12 and SDG 13

PS extensively used in packaging and consumer goods, is a major contributor to global plastic pollution, challenging waste management systems, and environmental preservation. This study is closely linked to Sustainable Development Goals (SDGs) 12 and 13, which focus on responsible consumption, production, and climate action [56–59]. As a non-biodegradable material, PS frequently ends up in landfills or oceans, leading to significant environmental harm [60–62]. Furthermore, the degradation and proper disposal of PS contribute to greenhouse gas emissions, exacerbating climate change. Effective management of PS waste through improved recycling and reduction strategies [63–65] directly supports SDG 12 by promoting sustainable production, minimizing waste, and encouraging the reutilization of materials. Advanced recycling methods, such as inverse polymerization and pyrolysis, offer significant potential in converting Waste PS (WPS) into valuable resources [66, 67]. These techniques not only reduce the environmental footprint of PS but also minimize the emissions of greenhouse gases associated with traditional waste disposal methods, aligning with SDG 13. Innovative recycling approaches not only help decrease plastic pollution but also support the principles of a circular economy, where waste is minimized, and resources are constantly reused. By adopting these sustainable practices, industries can play a crucial role in achieving these SDGs, particularly by reducing plastic pollution, conserving natural resources, and mitigating the adverse effects of climate change through lower greenhouse gas emissions. This integration of innovative waste management solutions into industrial processes is essential for promoting a more sustainable and resilient future for the country.

1.2 Synthesis of PS

PS is produced through the polymerization of the monomer styrene originally discovered in certain trees like *Liquidamber orientalis* around 200 years ago [68–70], as illustrated in Fig. 1. Styrene is primarily manufactured by dehydrogenating ethylbenzene, derived from ethylene and benzene through friedel–crafts catalysis employing aluminum chloride or a gas-phase process with zeolites [71–73]. An alternative method involves propylene oxide through an ethylbenzene hydroperoxide intermediate and styrene co-production. Styrene can be polymerized into PS through various mechanisms, including radical, anionic, cationic, and metallocene catalysis [74–76], using methods such as emulsion, suspension, bulk processes, and solution. Radical polymerization is the most familiar method, which can be induced thermally or with peroxides [60, 77–79]. It is believed that styrene forms a Diels–Alder adduct in thermal initiation, that reacts further to generate radicals that initiate polymerization [80–82]. The reaction follows first-order kinetics at high monomer concentrations, with termination occurring mainly via radical recombination or, less frequently, disproportionation [83–85].

GPPS can be described as a linear polyethylene chain with phenyl groups attached laterally along the chain. Among the isomers of PS, IPS, where all benzene rings are arranged on the identical side of the polymer chain, is not produced commercially at that mass [86, 87]. Due to its slow crystallization rate, its use in injection molding applications is limited [88–91]. SPS, with phenyl groups consecutively positioned on opposite sides of the polymer chain, shows a much faster crystallization rate compared to IPS [92–94]. However, APS, randomly characterized by scattered phenyl groups on both edges of the hydro-carbon, lacks crystallinity and is commercially the utmost substantial isomer of PS (Fig. 1). The C–C double bond in styrene can function as either an electron-donating or electron-withdrawing center [95–98]. Due to styrene's low polarity and the resonance stabilization of the growing PS species in the transition state, PS can be synthesized through various polymerization methods, including free radical, anionic, and cationic polymerization [99–103], as well as metal-catalyzed conditions (Fig. 2). The molecular structure of PS varies depending on the synthesis method used, resulting in linear, branched, star-shaped, comb-like, or dendritic forms [104–107]. For instance, in the free radical polymerization of styrene using benzoyl peroxide as an initiator, the process is initiated by the decomposition of benzoyl peroxide, generating free radicals that start the polymer chain reaction [108, 109]. Apart from peroxides,

Fig. 2 Common methods for PS synthesis

other initiators such as azo compounds, redox systems, and even thermal initiation without chemical initiators can be employed to start styrene polymerization.

After the formation of the benzoyloxy radical from benzoyl peroxide, it reacts with styrene to create an active center, which initiates the propagation step of the polymerization process [83, 110, 111]. The chain of polymers remains to grow through this propagation till termination occurs. As well as the ending of the polymer chain, chain transfer reactions can also happen, where the growing chain transfers to other components involved in the reaction of polymerization. The transference agents of the chain can be the creator, solvent, monomer, polymer, or else an additional chemical agent. While these transfer reactions do not influence the polymerization rate, they can reduce the average molecular weight of the polymer [112–115]. By using transfer agents such as mercaptans, the molecular mass of PS can be controlled. However, controlling the molecular weight, molecular weight distribution, and molecular structure of PS in free radical polymerization is challenging. This issue can be addressed by using iniferters substances that function as initiators, transfer agents, and termination agents. Iniferters react with rising macromolecular radicals to form transient species, thereby reducing termination through recombination and disproportionation. Moreover, the presence of iniferters increases the concentration of reactive polymer chains relative to free radicals, which results in a decreased polymerization rate [116].

Free-radical PS can be synthesized using various techniques such as bulk, solution, suspension, or emulsion polymerization. In bulk polymerization, pure styrene is used, resulting in a polymer with high clarity, but this method is not commercially preferred due to its poor control [117]. Solution polymerization involves dissolving styrene in a solvent, which allows for better temperature control but can lower the molecular weight and polymerization rate. Both bulk and solution methods can be performed in either batch or continuous processes, offering benefits such as more homogeneous products and lower volatile levels. Suspension polymerization involves dispersing the monomer in water along with stabilizers and initiators, making heat management and product recovery easier. Emulsion polymerization uses water as the carrier with emulsifying agents, producing very small particles, and is typically employed for making acrylonitrile–butadiene–styrene (ABS) copolymers [118, 119].

In current practices, GPPS is primarily manufactured through the continuous solution polymerization of styrene, utilizing heat removal via the evaporation of styrene and solvent. This controlled temperature process ensures high product quality [120, 121], characterized by narrow molecular weight distribution and excellent transparency [122]. Based on the production method, three distinct rankings of GPPS are available commercially: easy, medium flow, and high heat. Ps with both easy and medium flow are typically employed in injection molding applications, while high-heat PS is used for extrusion processes. The issue of plastic waste cannot be adequately addressed through landfilling or incineration [123], as these disposal methods are often too expensive and unsafe, particularly in developing countries [124, 125]. Moreover, incineration contributes to the release of greenhouse gases such as NO_x, SO_x, and CO_x, which drive climate change and also emit carcinogens [126, 127]. Chemical recycling of WPS into corresponding monomers or hydrocarbons has been explored by various researchers, however, this process is inefficient due to the lower cost of pure hydrocarbons and virgin monomers [29] compared to those obtained by recycling [128, 129]. Therefore, there is an urgent need to develop more effective procedures for recycling WPS. Although in developing countries, including India, there is very minute data on the disposal and potential uses of WPS, the country continues to consume a significant number of PS for various industrial and domestic purposes. For example, in India, the increasing generation of solid waste is a growing concern, and it is estimated that a substantial portion of this waste consists of WPS [130, 131]. This emphasizes the critical necessity for implementing efficient waste management strategies to mitigate the environmental impact of WPS.

2 Production of PS

PS was first synthesized in 1839 by German chemist Eduard Simon [132, 133] through the distillation or pyrolysis of liquid storax (Fig. 2). The industrial-scale production of styrene began in 1931, initiated by IG Farben industries in Germany and soon after by the Dow chemical company in the United States [134, 135]. The catalytic dehydrogenation of ethylbenzene is employed by both of these companies, a process that remains the primary method of styrene production today. This section will first discuss the polymerization process of styrene into PS, followed by an overview of how PS is processed into various finished products.

2.1 Polymerization process

High molecular weight vinyl aromatic polymers, particularly those with a weight average molar mass (Mw) exceeding 300,000, are typically produced via anionic polymerisation [136, 137] rather than free radical polymerization [138]. This preference arises from the slow polymerization rates associated with the free radical polymerisation of styrene, which makes it challenging to attain a high mass of molecules. However, anionic polymerization involves the use of expensive initiators, such as organolithium compounds, and often results in discolored products due to residual lithium salts [139, 140]. As a consequence, a high mass of molecule vinyl aromatic polymers is increasingly produced using free-radical polymerization, which avoids these issues and allows for the creation of bimodal polymer compositions. In the case of styrene, using an organolithium initiator in a hydrocarbon or ether solvent typically produces an atactic polymer with a narrow molecular weight distribution [141, 142].

High molecular mass vinyl aromatic polymers are typically manufactured in the presence of a soluble organic acid with a pKa between 0.5 and 2.5 by free radical polymerisation. However, during this process, the acid does not bind to the polymer and may migrate during the polymer's use, potentially leading to erosion of mold surfaces [143, 144]. Styrene has been polymerised using all four distinct polymerization methods: free-radical, cationic, anionic, and ziegler (Fig. 2). Among these, free-radical polymerisation is the most commercially prevalent [145], producing atactic PS widely used in packaging and durable products [146, 147]. This method is applied industrially through two processes: continuous (bulk or mass) and suspension polymerisation. Several parameters, for example, the type and quantity of initiator, surfactant, experimental conditions like polymerization temperature, and polymerization technique can be varied to synthesize and design PS with desired properties [148, 149]. Taued and Kosmella investigated the polymerization of styrene at temperatures from ambient to 373 K with different emulsifier regimes including persulphate, 2,2-azoisobutyronitrile, and surface-active initiators. Initiation energies displayed a robust dependence on the initiator emulsifier systems used [150]. McIntyre et al. demonstrated that mono-dispersed PS with molecular weights up to 44×10^6 can be synthesized using tetrahydrofuran as the solvent and polystyryllithium in benzene as the initiator [151].

Various practices take part in the polymerization of monomer styrene. For example, atom transfer radical polymerization [108, 152] was carried out using bromo acetylated syndiotactic PS as a macroinitiator, with copper bromide and PMDTA (N-Pentamethyl diethylenetriamine) acting as catalysts [153, 154]. Also, ultrasonically initiated mini-emulsion polymerisation of styrene in the presence of $FeSO_4$ nanoparticles was carried out [155]. The assimilation of $FeSO_4$ nanoparticles significantly accelerated the process of polymerisation, which is attributed to the increased aural intensity and the reaction of Fe^{2+} with H_2O_2 to generate hydroxyl radicals [156], in that way increasing the concentration of free radicals. Further, augmented concentration of co-surfactant too contributed towards the higher polymerization rate. Alternative methods for producing PS involve using the blatter radical and its derivatives in mediated stable free radical polymerization [157]. This approach opens up new options for radicals designing and developing to improve the polymerization process.

2.2 Processing of PS

PS processing is a well-established field, with continuing developments focusing on additives that modify the properties of the final product [158]. While the most commercially dominant form is GPPS, there are some further kinds of PS: EPS, HIPS, and SPS. These are produced via various polymerization processes. GPPS and HIPS are manufactured using continuous bulk mass polymerisation techniques [159]. In the early 1940s, Dow developed an extrusion process for PS to create a closed-cell foam resistant to moisture, which led to the creation of styrofoam products [132, 160]. EPS foam beads are made by a two-step process, where a blowing agent is introduced into pre-prepared resin beads in an aquatic suspension [29]. Also, EPS can be produced in a one-step process by directly impregnating the blowing agent into PS beads formed through suspension polymerisation [161, 162].

EPS products have low mechanical strength, which has led to the development of polyolefin bead foams to address this issue [163]. While polyolefin beads offer superior resilience, they do not retain the blowing agent used in their production. As a result, polyester foam was developed. This resilient polyester foam is both expandable and moldable, created from an amorphous polyester resin using a combination of a high-solubility compound and a low-permeability compound. The high-solubility compound allows for significant expansion, while the low-permeability compound provides additional expandability. Current research in the processing of injection foam employing the styrene-maleic anhydride co-polymer has given rise to a diverse and major upsurge in the performance of foam for food packing applications. Dow chemical

company's technology has enabled the synthesis of ethylene-styrene interpolymers. The features of these interpolymers vary considerably reliant on the copolymer composition.

3 Properties of PS

PS is a versatile, amorphous polymer with a molecular weight typically ranging from 100,000 to 400,000 [164–166]. It is known for its low density, high transparency, brightness, and lack of color, which, combined with low shrinkage and ease of fabrication, make it useful widely [167–169]. Below its glass transition temperature (T_g), PS behaves elastically according to Hooke's law, especially under low stress and strain conditions [170, 171]. The material's mechanical properties, such as tensile and flexural strength, are strongly influenced by its molecular weight. When the molecular weight drops below about 70,000, PS loses its structural integrity [31, 172]. Conversely, as the molecular weight increases, so do the tensile strength and elongation at break, until these properties plateau at a certain point. Under high impact or distortion, PS tends to form craze microvoids filled with oriented fibrils which help to dissipate impact energy [173, 174] across a larger area and play a critical role in the material's tensile strength. In contrast, rubber-modified PS, which has rubber particles grafted onto the polymer matrix, exhibits significantly greater toughness. This enhanced toughness relies on the strong adhesion between the rubber particles and the PS matrix [175, 176], which depends on the grafting process's efficiency.

In the above glass transition temperature (Tg), PS displays viscoelastic melt behavior. Two primary factors influence its elastic and viscous properties: the molecular weight and the molecular weight distribution of the polymer [177–179]. For monodisperse PS, the zero-shear viscosity is proportional to $M^{3,4}$. Although the ratio of weight-average molecular weight to number-average molecular weight has a minimal effect on viscosity, it significantly impacts steady-state shear compliance. Specifically, incorporating small amounts of high molecular weight PS into medium molecular weight PS can enhance both elasticity and melt strength [180, 181]. Summary of properties of PS is given in Table 1.

3.1 Degradation of PS

PS is a highly durable thermoplastic commonly thought to be non-biodegradable. Whereas PS does undergo biodegradation, the process is extremely slow in natural environments [68, 182, 183], leading to its persistence as solid waste for extended periods. It has been observed in a study that in cultivated soils teeming with fungi, microbes, and invertebrates, PS degradation was less than 1% after 90 days [184, 185] with no noteworthy increase in the degradation rate beyond this period [186]. In the same way, another study found that a sheet of PS was buried in soil for 32 years and displayed no signs of degradation [187–189]. The aquaphobic nature of most thermoplastics contributes to their resistance to hydrolysis. The molecular structure of plastics impacts their surface hydrophobicity [190, 191], which in turn affects the ease with which microorganisms can attach. Due to their high molecular masses and insolubility in water, microorganisms

Table 1 Summary of PS properties

Properties	Description
Chemical Formula	$(C_8H_8)_n$
Recycling Code	6
Appearance	Transparent, glassy solid; can be colored with pigments
Density	1.04–1.06 g/cm^3
T_g	Approx. 95 °C–105 °C
Melting Point	No true melting point; softens at 240 °C–260 °C
Thermal Conductivity	Low, 0.033 W/m K at 25 °C
Electrical Insulation	Excellent insulating properties; resistivity around 10^{16} ohm·cm
Water Absorption	Very low, typically less than 0.1%
Mechanical Strength	Good rigidity; tensile strength around 35–60 Mpa
Impact Resistance	Comparatively low, especially in brittle form; can be improved with additives (e.g., HIPS)
Chemical Resistance	Resistance to many acids, alkalis, and salts, but can be attacked by organic solvents like acetone and benzene
Processing Methods	Injection molding, extrusion, blow molding, and thermoforming

are incapable of transporting these plastics into their cells for metabolism. Biological degradation can begin outside the microbial cell through the secretion of extracellular enzymes [192]. However, these enzymes are too large to enter deeply into the polymer, so they work on the surface by breaking down the polymer chains through the mechanisms of hydrolysis [193]. The process is further facilitated by the formation of functional groups within the polymer chain.

In the manufacturing of thermoplastics, several additives are employed to increase the material's durability and functionality [194, 195]. These additives include antioxidants, flame retardants, processing lubricants, and stabilizers, each serving a specific purpose to improve the entire quality and lifespan of the resin [196, 197]. Antioxidants, such as Bisphenol A, are vital in preventing the oxidation of thermoplastics. Oxidation can lead to degradation of the material, so antioxidants help maintain the resin's integrity over time. Along with antioxidants, flame retardants are added to reduce the flammability of the plastic, making it safer for use in a variety of applications [198, 199]. Processing lubricants facilitate the manufacturing process by reducing friction and wear between machinery parts, ensuring a smoother production cycle and higher-quality end products. Stabilizers are particularly important for extending the service life of plastics, especially when used in outdoor environments. These stabilizers work by mitigating the effects of environmental factors such as UV radiation, high temperatures, and extended exposure to sunlight. UV stabilizers, also known as light absorbers, play a key role in protecting thermoplastics from UV damage [200]. They prevent issues such as discoloration, cracking, and brittleness that can occur due to sun exposure. Common UV stabilizers include benzophenones, hindered amines, and benzotriazole, all of which absorb or block harmful UV radiation, thereby preserving the plastic's desirable physical properties [201, 202].

To these traditional additives, recent advancements have introduced innovative materials like silver nanoparticles. Silver nanoparticles are utilized as antimicrobial agents in plastic food packaging, enhancing the shelf life of food products [203]. They achieve this by targeting microbial cells, waning their membranes, and interrupting enzymes responsible for nutrient transport, effectively preventing bacterial growth. Further additives, such as dyes and pigments, are employed to improve the aesthetic properties of plastics [204, 205]. These colorants not only enhance the visual appeal but also contribute to the functionality of the plastic by providing various kinds and finishes [206]. Broad usage of these additives in thermoplastics ensures that the material maintains its desired properties, performance, and safety across various applications and environments. The increasing demand for plastic has led to a rise in plastic waste, creating challenges in waste management due to limited landfill space, the threats of incineration, and rising disposal costs. Strategies for degrading PS using both pure microbial strains and complex microbial communities have shown that while PS can be biodegraded, the process is slow. Biodegradation of PS in landfills could help save space and recover energy through biogas [207, 208]. Another research area is identifying additives that accelerate PS biodegradation by promoting its reaction with atmospheric oxygen, such as metal salts and polymerizations [209, 210]. While these additives facilitate the process, their interactions and effects on biodegradation rates are not fully understood.

4 Applications of PS in the present scenario

PS is utilized in an extensive diversity of applications, including packing, manufactured goods, building materials, medical devices, and even arts and crafts [211]. PS is predominantly utilized in four main product categories: GPPS, HIPS, PS foam, and EPS foam [212]. The diverse applications of these products are outlined in Table 2. Here, we provide a brief description of some of these uses. PS vessels are usually used in the packaging industries and food services. Fresh food items can be well-preserved successfully when packed in PS. Further, electrical items are often packed in cases with PS support resources that provide insulation and protection during shipping. XPS (Extruded PS), commonly recognized as

Table 2 Important applications of PS

PS Types	Applications
GPPS	Disposable service wares (e.g. baskets, cookie trays, cupcake domes, cutlery, bowls, platters, and cups)
EPS	Cups, coolers (e.g. fish boxes), and isolated panel systems
HIPS	Cold drink cups, lids, plates (one-time use and reusable), stirrers, and yogurt containers
PS Foam	Poultry trays, hot and cold drink cups, use and throw plates, school lunches, other food serving trays, and foam sheet products (such as trays for eggs, fruits, and vegetables)

low-density styrofoam, along with EPS, is widely used in the manufacturing of trays, crockeries, boxes, takeaway food containers, coffee cups, and protective packaging for equipment or instruments [213].

PS is widely utilized in the production of various household appliances, including blenders, a.c, freezers, ovens, microwaves, vacuum cleaners, and also in disposable plastic cutlery [214]. Its lightweight nature and ability to provide a smooth, pristine finish make it a preferred material for these applications. It is used in the manufacture of pantry and restroom accessories, orchard tools, and toys. According to Inagaki and Kiuchi PS finds application in resin-molded articles like TV cabinets and cassette shells [215]. Ramesh noted its use in the production of soft-shell cycle helmets, as housing cameras, and in the inter-connected process to create extruded sheets and bun foams for foot gear components [216].

Further PS usages consist of construction such as insulating ceilings, fortifications, and floorings. It is also used to soundproof building walls [217, 218]. In addition, PS resins play a vital role in the making of illumination and drainage system fixtures, as well as panels [219]. In laboratories, it is used to manufacture essential things such as petri plates, tissue culture test trays, test tubes, analytical equipment, and plastic storage containers [220]. While, XPS is popular in arts and crafts projects, including architectural models, candle holders, and Christmas tree ornaments [29].

5 Common sources of PS waste

PS waste is generated from a wide range of industries, primarily due to its versatile use in packaging, consumer products, and insulation materials (Table 3). The packaging industry is one of the largest contributors, generating waste through disposable food containers, cups, plates, and protective packaging materials for electronics and delicate items [221]. The food and beverage industry adds to this with the use of PS in takeaway containers, disposable cutlery, and trays. In the construction industry, EPS and Extruded PS (XPS) are widely generated as insulation materials, contributing to waste during construction, demolition, or renovation activities [222]. The consumer goods sector produces PS waste from products like CD and DVD cases, disposable razors, and other household items. The automotive industry also plays a role in generating PS waste, as they use PS for parts, insulation, and packaging. In the electronics industry [223], PS is used for the protective packaging of devices and appliances. The medical and pharmaceutical sectors contribute through disposable laboratory items such as Petri dishes, test tubes, and pipet tips. Further, the retail industry uses PS foam for displays and packaging for delicate goods [224]. Together, these industries contribute significantly to WPS, posing challenges for sustainable waste management and recycling efforts.

6 Practices for disposal of PS

PS has remarkable stability and resistance to decomposition from its benzene rings and only strong c–c bonds [225]. This perseverance in the environment leads to various ecological impacts, such as visual pollution, wildlife entanglement that impairs feeding, and risks of strangulation and ingestion [226]. Also, PS releases carcinogens i.e., Poly Aromatic Hydrocarbons (PAHs). As their short volatility and poor biodegradability, PAHs can accumulate in living organisms once they enter the body [10]. Developed countries frequently manage PS waste through landfilling, incineration, or waste minimization processes to mitigate environmental and public health risks. In contrast, developing nations tend to use exposed burning and landfilling's for disposal. Due to PS's low density, it occupies significant space, reducing landfill

Table 3 Sources of WPS from various industries

Industries	PS Use	Waste sources
Packaging	Food containers, cups, plates, protective packaging	Disposable items, packaging materials
Food and Beverage	Takeaway containers, disposable cutlery, trays	Single-use food service items
Construction	EPS, XPS insulation materials	Building, destruction, renovations
Consumer Goods	CD or DVD cases, disposable razors, household items	Discarded consumer products
Automotive	Car parts, insulation, protective packaging	Automotive components
Electronics	Protective packaging for electronics	Packaging of electronic devices
Medical and Pharmaceutical	Disposable lab items like Petri dishes, test tubes	Laboratory disposables
Retail	For display purposes, packaging for fragile goods	Displays, protective packaging

capacity and lifespan. Further, landfilling represents an unexploited prospect to recover valued energy resources from WPS [60]. Incineration is the most common disposition method in developing countries, where high-heat incineration predominantly gives rise to Co_2, H_2O, volatile hydrocarbons [227], and heat. In comparison incineration of landfilling, at around 900° Celsius significantly reduces the material volume, producing about solids (35%) and gases (65%), mainly in the form of soot [228, 229]. The high calorific value of PS allows the high temperature generated through incineration to be harnessed for thermal and hydroelectric energy production. However, incineration has faced criticism due to producing an intricate mix of PAHs, especially at lower temperatures, such as 800–900 °C. Exposure to PAHs through skin interaction, swallowing, or breath can lead to genotoxic and oncogenic effects. PAHs in the body are metabolized to generate spontaneous power products like diol epoxides [230, 231]. These metabolites can either be expelled or else bind to DNA, creating PAH-DNA adducts, which can trigger the inception of carcinogenesis. Moreover, in the presence of light, PAHs have been stated to cause single-strand DNA breaks by the oxidation of DNA bases. The amount and types of PAHs produced during incineration are influenced by both the incineration temperature and the dimensions of the PS feed. Investigating the impact of PS feed size (ranging from 100 to 300 µm) and incineration temperatures (800, 900, 1000, and 1200 °C) on PAH production generated two important insights for the efficient operation of incinerators. Primarily, as the temperature rose from 800 to 1200 °C, there was an exponential reduction in both the total number of PAH species and the overall PAHs mass yield, with almost no PAHs detected at 1200 °C. Secondly, the study found that reduced-sized elements (104 to 175 µm in diameter) produced a PAH mass yield nearly seven times lower than that of larger elements. In total, the findings from these studies suggest that at higher temperatures and with smaller particle sizes more effective ignition of PS is achieved. However, the distribution of PAHs between the gaseous and solid phases was found to be unaffected by temperature [232–234].

Due to the public well-being hazards related to PAHs, various nations are implementing approaches to minimize their impact, focusing on reduction, recycling, reuse, and recovery. However, from an environmental point of view, paper packing consumes a greater carbon footprint as compared to PS. In a study, PS poses less environmental impact than paper because, in landfills, paper decomposes and releases methane (CH_4) and carbon dioxide (CO_2) in a 2:1 ratio [60]. Both CH_4 and CO_2 are noteworthy contributors to worldwide climate change, with CH_4 being around twenty times more effective than CO_2 as a greenhouse gas [235–237].

7 Reusability/Recycling of WPS

Recycling or Reusability involves converting post-consumer waste into value-added products, making it one of the most effective waste management approaches. The importance of recycling lies in its ability to preserve natural resources, as highlighted by Allwood who point out that recycling reduces the need for virgin materials like wood, oil, minerals, and petroleum products [238]. In some studies, recycling plays a crucial role in environmental protection by reducing pollution not only from waste but also from taking out and processing raw materials [239]. This process also helps in lessening the emission of greenhouse gases, a substantial contributor to environmental change. Reusability also addresses the critical issue of waste accumulation by dropping the volume of leftover waste sent to landfills and dumps, which in turn lowers the risks of land and water pollution and supports land conservation efforts [240, 241]. Beyond environmental benefits, reutilization is driving economic growth by creating new opportunities in sectors such as construction, manufacturing, and transportation [242]. It also provides financial benefits by cutting down the costs linked with the extraction and processing of raw materials, making it a key factor in sustainable development and resource management. WPS, particularly from packaging materials like food containers, cups, protective packaging, and insulation, can be efficiently reused and recycled. One of the most common forms of WPS that can be reutilized is EPS. Direct recycling involves creating novel products directly as of WPS, whereas indirect or secondary recycling includes modifying the WPS into new value-added products [243]. Common examples of recycling methods include catalytic cracking or pyrolysis to generate fuels, volatile gases, and goods such as ion exchange resins [244, 245], construction materials, and styrene [246]. Four common methods for treating solid plastic waste are re-extrusion, mechanical, chemical, and energy recovery. Advanced thermo-chemical treatments primarily produce fuels, petrochemical feedstocks, or recycled energy through the use of catalytic agents [247] such as ZSM-5 zeolite and alumina/silica. For example, Zhang et al. demonstrated that barium oxide can selectively degrade PS into monomer styrene and its dimeric form [248].

7.1 Production of monomer styrene through polymer degradation

The use of solid acids and bases for chemical recycling into monomers has been explored extensively. However, acids face the challenge of quickly becoming deactivated due to carbon accumulation on their surfaces. BaO is effective for degrading PS into styrene. Catalyzed pyrolysis of PS produces a high yield (more than 80%) of monomer styrene [249], whereas uncatalyzed pyrolysis yields less styrene and produces a mixture of scented hydrocarbons, with toluene, ethylbenzene, and polycyclic scented hydrocarbons. In a study by Sun et al. a technique for depolymerizing PS into its monomers was established using a platinum or rhodium (Rh) catalyst mixed with cerium and reinforced on silica in a constant method. In this setup, shredded particles of WPS were inserted into a reactor where they interacted with the heated catalytic agent, resulting in their pyrolysis into monomer styrene. The warmth generated by the reaction was adequate to sustain the reactor's high temperatures, rejecting the need for additional energy input. Besides styrene, the process also produced benzene, toluene, ethylbenzene, and small PS oligomers. The process attained a styrene monomer yield of about 80%. The cost-effectiveness of this method depends upon factors such as the sorting, cleaning, and transportation of the waste plastic [250]. Lilac and Lee investigated the degradation of PS using supercritical water, achieving a monomer styrene selectivity of 71%, while selectivity for benzene, toluene, and other hydrocarbons was significantly lesser. PS can be inserted into the reactor in either rigid or fluid form [251]. Further, in an investigation by Karmore and Madras, they examined PS depolymerization using Lewi's acids such as $FeCl_3$, boron trifluoride etherate, $AlCl_3$ (Aluminum chloride), and tin chloride at temperatures ranging from 75 to 125 °C, with $AlCl_3$ yielding the best outcomes [252].

7.2 Green recycling of EPS waste into nano and microparticles

In a study by Mangalara and Varughese a novel eco-friendly method has been developed for converting EPS waste into value-added products using an emulsification-diffusion technique. This approach employs d-limonene, a green solvent, to dissolve EPS waste and utilizes water as the non-solvent for the recovery of PS particles, with biodegradable polyvinyl alcohol (PVA) helping as the emulsifier. The size of the resulting PS particles can be controlled by adjusting process parameters such as PVA concentration, PS concentration, and homogenization pace. The study finds that the yield of PS particles is significantly influenced by the concentration of PVA, with optimal yields achieved at a 5 wt% concentration. Larger, micron-sized particles are produced at higher PS concentrations and lower homogenization speeds. In addition, the recovery of the solvent is integrated into the process, increasing its sustainability. This method provides a sustainable and green approach to recycling and recovering WPS, highlighting the potential for environmentally friendly alternatives in waste management practices [253].

7.3 Reprocessing of EPS and HIPS

In a study by Noguchi et al. they introduced an innovative prototype manufacturing system aimed at reprocessing EPS utilizing d-limonene (an organic natural solvent derived from citrus peels). This solvent effectively dissolves and reduces the volume of EPS, significantly decreasing the material and making it easier to handle and process. The recycled PS obtained through this technique is reported to exhibit mechanical properties closely matching those of virgin PS [254]. The study suggests that the recycled material can be employed in applications that traditionally require the use of new, unprocessed PS, thereby offering a sustainable alternative without compromising on quality or performance.

7.3.1 High-value goods

a) *Production of nanofibers PS*: Nanofibers, with their high surface-to-volume ratio, are beneficial for catalysis and the production of nanoporous filters, which are used in air and water purification systems. Nonporous filters made from non-woven PS can be produced using WPS. For example, electrospun PS nanofibers can be created from regenerated WPS using solvents like tetrahydrofuran and dimethyl formamide [255, 256]. These nanofibers are then crosslinked with sulfuric acid and modified with ethylene diamine to produce ion exchange fibers, which are useful in rapid water treatment and can serve as supports for catalytically active metal ions. Further, electrospun nanocomposites made by incorporating cellulose nano-whiskers into PS nanofibers exhibit special structural features that provide novel belongings and functions [257]. Electrospun PS nanofibers have a broad range of applications, including practice in detectors, protective clothing, purification, drug delivery, wound bandage, and tissue scaffolding. These nanofibers can also stabilize volatile substance's cyclodextrin enclosure complexes, such as menthol, at high temperatures. Also,

when combined with inclusion complexes such as 1-methyl cyclopropane, these nanofibers can inhibit ethylene-induced physical changes in many agronomic products, thereby extending their longevity [258].

b) Generation of super absorbers: WPS can be transformed into high-valued items like super-adsorbents. Safi demonstrated the development of a WPS-acrylic acid-bentonite super-absorbent nanocomposite using emulsion polymerization [38]. The nanocomposite, with 3% bentonite, demonstrated the maximum water porosity, achieving 500 g/g in distilled water and 49 g/g in a 0.9% NaCl solution. These super-adsorbent nanocomposites have potential applications in products like diapers and could also be used to remedy contaminated groundwater [259]. However, evaluating the risks associated with remaining styrene monomers and further PAHs in such products is crucial.

c) *Paint production*: WPS can be efficiently used as a binder in paint production. Some research has shown that petrol is the most suitable solvent for dissolving WPS, resulting in enhanced elongation at break, with a maximum solubility of < 20% (w/v). When combined with other binders, styrene homopolymers are particularly useful in applications like galvanized steel dirt primers for anti-corrosive, paint for mineral substrates, and bronze glosses [260, 261]. Styrenated alkyd resin coatings offer advantages such as faster drying times, improved resistance to light, water, and chemicals, and reduced yellowing of films [262, 263]. This highlights the importance of the paint industry's utilization of WPS as a main component in numerous applications.

d) *Production of ion exchangers and flocculants*: In a study by Inagaki et al. they demonstrated the redevelopment of WPS through sulfonation, which transformed it into a polymeric flocculant with better belongings than traditional polyacrylamide. They also found that combining sulfonated WPS with polyacrylamide created a synergistic effect. This process is attractive due to the high expense and partial accessibility of water treatment chemicals [264]. In addition, sulfonated WPS has been utilized to adsorb substantial metals for example Cd^{2+}, Pb^{2+}, and Zn^{2+} from water, showing noteworthy removal abilities, however not as effective as some biosorbents. PS has been utilized to create innovative ion-exchange resins with significant industrial uses, including water and wastewater treatment, soil remediation, and the purification of industrial products like hormones [265, 266] and enzymes by bio-separation and chromatography [267]. Current developments include inert-linked sulfonated PS ion exchange resins that offer high capacity and rapid ion exchange, making them effective at removing divalent ions such as calcium, magnesium, nickel, and copper from water. These PS-based exchangers provide a cost-effective solution for water treatment, especially in regions with limited access to hygienic drinking water.

Ion-exchange membranes can be transformed by WPS for the purification of H_2O_2 through specific chemical modifications [268, 269]. One approach involves creating anion exchange membranes by first cross-linking the PS and then modifying it with gas-phase nitration [270], and subsequently carrying out amination using hydrazine hydrate [271]. These membranes are then treated with di-chloroethane and tri-ethylamine to acquaint with quaternary ammonium groups, which boost their ability to eliminate chloride ions through electrodialysis [272]. One more innovative use of WPS is in the expansion of hyper-crosslinked PS sorbents, which have shown efficiency in removing phenol from water. This application is mainly valuable for treating industrial effluents polluted by phenol [273]. Ion-exchange membranes can also be adapted for mineral processing by incorporating specific selective chemical groups [274]. Despite these promising applications, it is important to manage a detailed benefit of cost evaluation to ensure the monetary feasibility of these procedures. This evaluation should contemplate the costs associated with collecting and cleaning the foam waste, as well as the expenses related to its reprocessing into functional materials.

7.3.2 *Nanocomposites of PS* with *polymer clay*

Nanocomposites with polymer–clay show enhanced properties compared to pure polymers, including improved modulus, ductile strength, and heat distortion temperatures, along with reduced sensitivity of water and gas penetrability [275, 276]. WPS can be recycled into these nanocomposites, which are valuable for applications in packaging, toy manufacturing, and electronic casings due to their superior strength and consistency. Recycling processes can affect the mechanical properties of materials like ABS and HIPS [277, 278], potentially leading to reduced strain and impact strength due to chain length shortening. However, incorporating montmorillonite nano clay into recycled WPS can significantly enhance these properties. HIPS is noted for its resilience in mechanical recycling, maintaining its properties through multiple processing cycles [279]. Moreover, green recycling methods using natural solvents for WPS do not degrade the polymer and can enable its conversion into various products, including nanocomposite membranes, electrospun carbon nanofibers, and other new items.

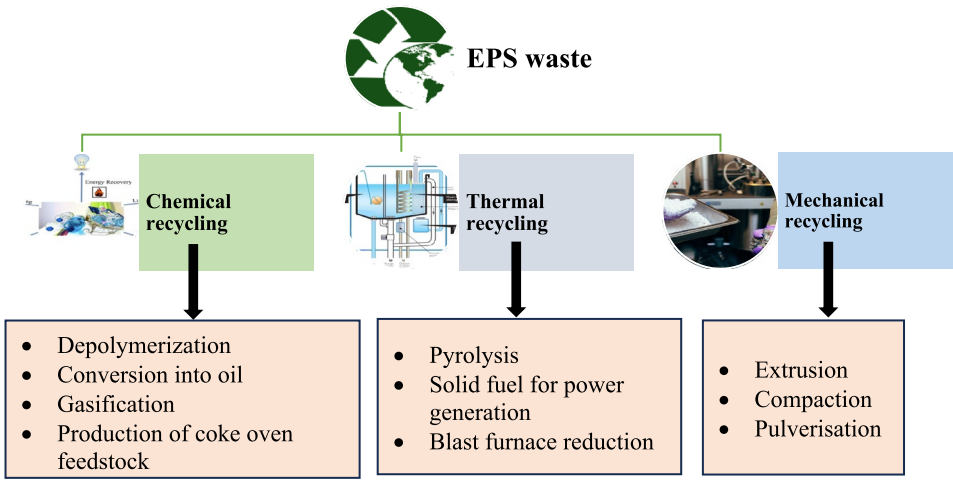


Fig. 3 Various methodologies for EPS waste recycling

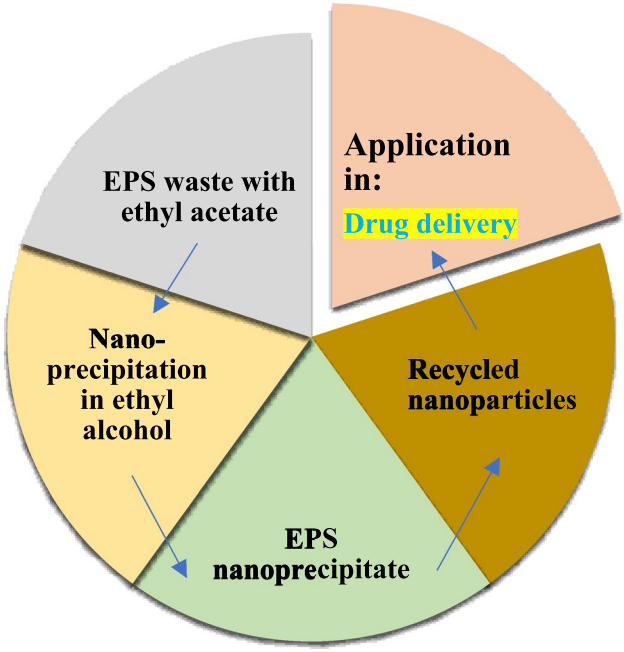


Fig. 4 Recycling of EPS waste into nanoparticles for potential applications

8 Recycling of EPS waste

PS is commonly available in three forms: solid PS, EPS, and HIPS, which are reinforced with polybutadiene [280, 281]. All three types are extensively used in packaging applications. While solid PS can be easily recycled mechanically into useful products, EPS frequently requires more complex processes, such as solvent-based or mechanical methods [282], to de-foam and reduce its volume before further processing. EPS is a highly versatile and lightweight material commonly used in packaging, thermal insulation, and protective padding due to its excellent cushioning properties and thermal resistance. However, despite its many practical applications, EPS poses significant environmental challenges because it is non-biodegradable and can persist in the environment for hundreds of years if not managed properly [283]. This persistence leads to environmental pollution, particularly in marine ecosystems, where it can cause harm to wildlife. To mitigate these issues, recycling and safe disposal of EPS are critical. There are three main methods for recycling EPS: (a) Chemical recycling; (b) Mechanical recycling, and (c) Thermal recycling. The different pathways available for recycling EPS waste are shown in Fig. 3.

8.1 Process of chemical recycling

The chemical recycling of EPS involves depolymerizing the material to recover styrene monomer, with catalytic degradation being a noteworthy method [284, 285]. EPS can be chemically converted into rigid PS, which broadens its utility in various applications. One approach involves dissolving EPS in ethyl acetate to form a solution, which is then precipitated in ethyl alcohol to produce EPS nanoparticles [286, 287]. These nanoparticles have potential applications in drug delivery [288, 289] and coatings (Fig. 4). Another innovative approach dissolves EPS in styrene monomer within a closed-loop system. This method employs suspension polymerization of the EPS-styrene solution, incorporating the styrene monomer throughout the mass, which streamlines processing by eliminating the need for solvent and polymer separation [246]. This process is advantageous compared to traditional chemical recycling methods, offering a more efficient and sustainable route for EPS recycling.

According to earlier studies, including those by Koji et al. demonstrated that PS foam could be reduced and reclaimed into styrene by catalytic decomposition using basic metal oxides combined with an inert blowing agent. They investigated the effectiveness of basic oxides such as magnesium oxide (MgO), sodium oxide (Na_2O), and calcium oxide (CaO), with CaO being identified as the most desirable catalyst [290]. Their experiments involved decomposing EPS to styrene within a temperature range of 300–450 °C in a non-oxidizing atmosphere. Siyal et al. also investigated the chemical recycling of EPS waste by changing it into a new PS-hydrazone (PSH) surface, which was then used to treat industrial wastewater contaminated with phenol [291]. Their method involved the acetylation of PS followed by condensation with phenylhydrazine, and the resulting PSH surface effectively treated phenol-laden industrial wastewater, highlighting a promising approach to managing EPS waste and addressing phenol contamination in water. Further, Abdel et al. reported using dialkyl carbonate solvents in their method, which involved filtration to remove insoluble components [292]. Subsequently, they employed a non-solvent or a mixture of non-solvents to selectively precipitate PS, which was then isolated, dried, and extruded. This approach allowed for the recovery of pure PS without altering its intrinsic properties.

8.1.1 Benefits or limitations of chemical recycling

Chemical recycling, mostly through depolymerization, offers high recycling efficiency by breaking down EPS waste into its monomer, styrene. It has lower energy requirements than thermal and mechanical recycling methods and opens up opportunities for various post-consumer applications of EPS waste. However, this method also has limitations, including the process's complexity, which requires specialized expertise and experience, hindering the commercialization of findings. Further, the chemicals involved in chemical recycling can be expensive and eco-friendly, potentially releasing toxic materials. Even when using natural-based solvents like d-limonene, the cost remains higher [293].

8.2 Process of mechanical recycling

The mechanical recycling of EPS waste typically begins with sorting, washing, and drying, followed by size reduction into flakes or granules using a milling machine [294]. After this, further treatments like solvent treatment, extruder (for blending in wood-plastic composites), or direct blending can be performed. The milled EPS powder can also serve as a fine aggregate filler in various plastic processing applications [295]. Pulverization, another method for recycling EPS, involves grinding the material into smaller pieces, which can then be remolded into new EPS products or used as a filler in other products [296, 297]. This method effectively reduces waste, conserves resources, and minimizes the environmental impact associated with EPS production and disposal. Modern plastic sorting techniques combine automated and manual operations, using technologies such as Near-Infrared (NIR) spectroscopy to identify polymer types, along with optical color sorting for separating plastics into clear and colored fractions [298]. Other sorting technologies include X-ray, density-based methods, electrostatics, melting point differentiation, selective dissolution, and manual sorting. EPS waste can be well recycled by several mechanical methods, such as milling, granulation, compression, extrusion, and injection molding [299]. Mechanical recycling using a milling machine helps reduce the size of EPS without significant loss of properties [300], and the resulting material can be used in applications such as lightweight concrete with compressed strength aligning with worldwide standards [301]. Granulation includes shredding EPS into tiny pieces and melting them to form beads, which can be further utilized in products like silicate spheres [302]. Compression techniques compact EPS waste into blocks or logs, which are then repurposed for applications like soundproofing or thermal insulation in building materials [303]. Also, extrusion processes melt EPS in an extruder to create plastic sheets or pellets, while injection

molding uses melted EPS to form new products [304]. Amongst the compactors used, screw compactors are favored over hydraulic ones due to their more competent pressure regulation. These mechanical recycling methods help in the sustainable management and reusing of EPS waste, contributing to environmental conservation.

8.2.1 Benefits or limitations of mechanical recycling

In comparison to chemical and thermal recycling, the mechanical method of recycling EPS has many advantages. It is eco-friendly since it does not emit volatile organic compounds or unpleasant smells related to petroleum-based solvents used in EPS recycling. Mechanical recycling conserves the integrity of EPS, unlike thermal methods that can degrade the material, making it an energy-efficient substitute [305]. It is also cost-effective due to its straightforward process, allowing for the production of various products like plastic lumber and EPS insulation. However, the main drawback of mechanical recycling is the potential for increased costs and lower quality of the final recycled product as compared to virgin materials [306]. In addition, the mechanical recycling of EPS is not always the most economical choice due to these quality concerns.

8.3 Process of thermal recycling

Thermal recycling is active in the management of waste to decompose waste materials or reduce their volume through heat application, aiming to convert waste into sustainable, reusable products and lessen landfill disposal. The main methods of thermal recycling include fluid catalytic cracking and pyrolysis. Fluid catalytic cracking involves controlled heating or cracking of waste at high temperatures and atmospheric pressure, using acid catalysts such as silica-alumina's or zeolites [307, 308]. A significant process utilizes a fluidized bed with sand to feed solid-state polymers. Pyrolysis, on the other side, is suitable for plastics that are difficult to depolymerize, such as polyethylene, polypropylene, PS, composites, and polyurethanes [309]. It operates by heating plastics at moderate temperatures (450–500 °C) in the absence of oxygen, breaking down polymers into smaller molecules [310], resulting in gassy, fluid, and firm char fractions. Pyrolysis offers significant advantages by enabling the treatment of contaminated plastics, without the need for extensive sorting, and it can transform large complex plastic molecules into smaller, valuable, hydrocarbons suitable for a range of applications [311]. However, its major drawbacks include high energy requirements and extended reaction times [312]. Many researchers have explored the use of pyrolysis processes for recycling EPS waste. For example, de Marco and Poletto examined the depolymerization of EPS via microwave-assisted pyrolysis using a carbon black catalyst. Their studies confirmed successful EPS pyrolysis at 400W over a fixed duration of 12 min, suggesting that depolymerization of EPS through microwave heating with carbonaceous catalysts offers a viable route for recycling [313]. Similarly, another study reported pyrolysis by microwave-assisted WPS using an activated carbon catalyst, achieving an optimum oil yield of 93% at 450W with a polymer-to-activated carbon ratio of 10:1, conducted at a temp. of 330 °C for 5.5 min [314].

The depolymerization method of EPS waste through microwave-assisted pyrolysis is familiar as an effective recycling method, with noteworthy results influenced by factors such as catalyst type, depolymerization time, and temperature [315]. In a study, it has been shown that graphene oxide catalysts offer superior thermal stability as compared to activated carbon, mainly due to their active heat transfer properties during the process of pyrolysis [316, 317]. Another study explored the impact of varying microwave power levels (300 W, 450 W, and 600 W) on the yield of EPS depolymerization, revealing an ideal conversion efficiency of 68.1% at 600 W using a smaller amount of graphite [281]. This indicates a positive correlation between increased temperature and yield, suggesting that higher temperatures can enhance the depolymerization process. Materials with high thermal conductivity, such as metals or carbonic compounds, show a dynamic role by efficiently absorbing and transporting heat to the polymer, so promoting better depolymerization outcomes. In total, the efficiency of EPS recycling through microwave-assisted pyrolysis is significantly affected by the interplay of temperature, catalyst type, and material properties. Commercializing direct pyrolysis of plastic presents challenges due to the heterogeneity of the plastic and the associated limitations in heat and mass transfer [318]. Catalysts are frequently used to lower the operating temperature, preventing the complete degradation of the plastics. A different method, the recycling method [244] involves heat treatment that significantly reduces the volume of EPS by approx. twenty times while increasing its density. This process, which they termed Modified-EPS (MEPS), also altered other properties such as thermal conductivity and compressive strength [319]. They achieved these results by heating the samples in an oven at various temperatures (110°–150°C), which allowed for volume reduction and increased thickness, making MEPS suitable for use in concrete applications [320].

8.3.1 Benefits or limitations of thermal recycling

Thermal recycling of EPS offers some advantages, including quick conversion into practical products and cost-effectiveness due to the absence of chemical requirements. Its adaptability allows for recycling various PS forms, and some processes even recover energy, which can be repurposed. However, this method also has notable drawbacks/limitations. It demands substantial energy input due to high-temperature requirements, making it costly and contributing to greenhouse gas emissions. Challenges in temperature regulation and maintaining the quality of the recycled material pose further limits to thermal recycling [321].

9 Applications of recycled WPS

Recycled WPS finds diverse applications across various industries (Fig. 5), due to its diverse properties. It can be used in the production of insulating materials, such as EPS boards, which are widely used in building construction for thermal insulation [322]. Recycled PS is also repurposed into packaging materials, including protective packaging for electronics, food containers, and disposable cutlery [323].

In the automotive industry, recycled PS can be used for manufacturing lightweight components, contributing to vehicle mass reduction and enhanced fuel efficacy [324, 325]. Innovations in recycling technologies, such as reverse polymerization and microwave-assisted pyrolysis, are expanding the potential applications of recycled PS by converting it into monomers or other valuable chemicals, which can then be used as raw materials in various industrial processes.

9.1 A new approach to PS waste in the field of chemiluminescence (CL)

Chemiluminescence is a process in which light is emitted as a result of a chemical reaction, and it has emerged as a powerful imaging technique at the interface of chemistry and biology for detecting specific biological processes, such as catalytic activities and reactive small-molecular flux [326, 327]. This process finds widespread applications in high-throughput assays, as well as in living-cell and live-animal imaging. There is a rising focus on developing new probes and systems with strong and prolonged emissions to achieve highly sensitive CL imaging. Scientists and researchers have explored several CL nanoprobes and substrates to enhance emission intensity, extend duration, and modify wavelengths. Important CL systems used for in-vivo imaging consist of luminol and its analogues, peroxyoxalate, lucigenin, imidazopyrazinone, and oxalyl chloride [328]. While non-enzymatic peroxalate-based CL reactions offer long emission times and specific reactions to hydrogen peroxide (H_2O_2), their strong hydrophobicity limits their application in aqueous or physiological environments [329, 330]. Water-dispersive peroxalate nanoparticles integrating peroxalate and

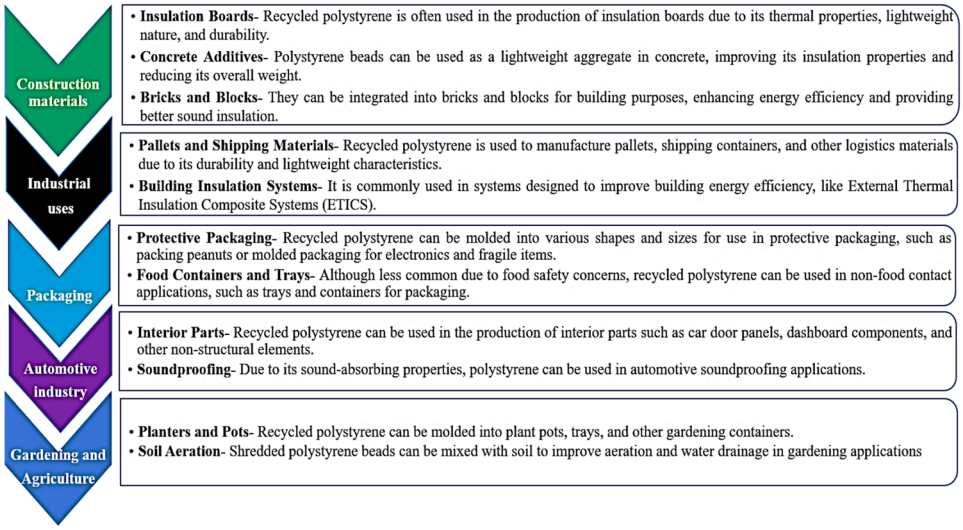


Fig. 5 Common applications of recycled WPS

hydrophobic fluorescent molecules have been synthesized to address this [331]. These nanoparticles act as nanoreactors for sensitive and selective in-vivo imaging of H_2O_2, operating through a chemically initiated electron-exchange luminescence mechanism [332].

Zhou et al. examined mimic-enzyme catalytic chemiluminescence (CL) polymer dots, specifically PS-hemin dots, composed of hemin, PS, and an amphiphilic polymer, which demonstrated a remarkable 6000-fold increase in CL emission compared to the direct reaction of hydrogen peroxide with CL substrates. This significant enhancement is attributed to a high density of hemin molecules within each nanoparticle, the high catalytic efficiency of the PS-hemin dots, and a slow-diffusion-controlled heterogeneous reaction that prolonged emission over more than 5 h. The PS-hemin dots demonstrated excellent stability, biocompatibility, and ultrasensitive response to hydrogen peroxide, making them highly suitable for bioassays and bioimaging. In their study, the authors conjugated PS-hemin dots with integrin-targeting ligand cyclic RGD (arginine-glycine-aspartic acid) to facilitate CL imaging of normal and cancerous cells by exploiting the specific interaction between integrin αVβ3 and cyclic RGD. Also, PS-hemin dots were employed for real-time imaging of reactive oxygen species (ROS) levels in normal and inflamed mice, highlighting their potential for in vivo applications. Compared to traditional enzyme-based CL probes and metal-based nanoparticles, PS-hemin dots offer superior stability, sensitivity, and biocompatibility, underscoring their promise as highly sensitive CL probes for targeted diagnostics, therapy, and advanced bioimaging applications [333].

Som et al. developed a sensitive CL method for arsenate detection in H_2O using a microfluidic device with PS beads for sorbent pre-concentration. The method is based on the formation of vanadomolybdoarsenate hetero polyacid (VMoAs-HPA), which ion-pairs with hexa-decyl-trimethyl ammonium bromide and sorbs onto the beads. Optimal concentrations of reagents were determined using a variable-size simplex optimization, and ethylene diamine tetra acetic acid (EDTA) was used to minimize interference from other metal ions. This approach significantly enhanced detection sensitivity and reduced interference from phosphate and chromate. The method demonstrated a linear response for arsenate concentrations from 1.0×10^{-7} to $5.0 \times 10^{-}5$ M, with a detection limit of 8.9×10^{-8} M, and a rapid analysis time of 5 min per run with good reproducibility (relative standard deviation of 5.9%, n = 9). It was effectively implemented in various water samples, including mineral, tap, and drinking water [334]. A clear overview of some more literature reviews is shown in Table 4.

CL, the release of light from chemical reactions, is widely used in biomedical applications, analytical chemistry, and environmental monitoring, due to its high sensitivity and low background interference. A new approach to take advantage of PS waste in the field of CL involves reusing or recycling the waste materials into purposeful components that enhance CL-based systems or processes. This approach comprises the functionalization of PS waste to develop reactive surfaces or particles that can improve CL signal intensity and stability, the application of modified PS as substrates or matrices in CL reactions, and the development of microfluidic devices or PS-derived materials for sensors in CL- based detection systems. These inventions not only address environmental concerns by reducing plastic waste but also make available cost-effective solutions and probable performance enhancements, such as increased signal-to-noise ratio and healthier analysis designs. Yet, challenges remain, together with ensuring the sustainability of recycled PS materials, scalability of processes, and compliance with governing and safety standards, specifically in biomedical applications.

The mechanism employed in this study relies on the chemiluminescent properties of the novel 1,2-dioxetane phosphate derivative (AMPDP) in conjunction with alkaline phosphatase as a catalytic enzyme. The AMPDP substrate is enzymatically cleaved by alkaline phosphatase to generate an unstable intermediate that decomposes to produce light. The emitted light is detected and quantified, providing a measure of the analyte concentration. The polystyrene beads (PSBs) and ferrite-coated particles serve as solid-phase carriers, enhancing the reaction's efficiency by providing a high surface area for enzyme–substrate interaction. This optimized setup reduces reaction time to 30 min and improves sensitivity significantly, enabling precise and rapid detection compared to traditional radioimmunoassay (RIA) and enzyme immunoassay (EIA) methods.

10 Challenges with WPS

The WPS recycling faces several challenges. Despite the well-established technology for PS recycling into numerous products, the process is hindered by transportation [247] issues due low density-volume ratio of the materials. Gathering, storing, and transferring PS over long distances is economically impractical. Further, a lack of incentives to advance recycling efforts is there because of inadequate investment in framework, logistical systems, and compaction equipment. PS has high constancy, attributed to its benzene rings and solo bonds of C–C, meaning that its breakdown or depolymerization needs energy-intensive conditions, including elevated heat and pressure [225]. However, a recent study indicates that

Table 4 An overview of chemiluminescent studies using PS as a base material

Study	Methodology	Main components	Mechanism	Applications	Refs.
Nishizono, et al.	The methodology involved developing a chemiluminescent enzyme immunoassay (CLEIA) using a novel 1,2-dioxetane phosphate derivative (AMPDP) and alkaline phosphatase. The assay, tested with PS Beads (PSBs) and ferrite-coated particles, reduced reaction time to 30 min. It achieved linear standard curves and improved sensitivity by two- to ten-fold compared to conventional RIA or EIA methods	*Tumor Markers*: Carcinoembryonic antigen (CEA), alpha-fetoprotein (AFP), CA19-9, CA125 - *Substrate*: AMPPD - *Labeling Enzyme*: Alkaline phosphatase - *Solid Phase*: PS beads, ferrite-coated particles	The study developed a chemiluminescent enzyme immunoassay (cLEIA) for detecting tumor markers using a novel chemiluminescent substrate	- *Quantification of Tumor Markers*: Useful for measuring levels of CEA, AFP, CA19-9, and CA125 - *Clinical Diagnosis*: Enhanced sensitivity for detecting tumor markers in serum samples - *Comparison with Other Assays*: Correlates well with RIA and EIA methods	[335]
Lee et al.	Nanoparticles from peroxalate esters and fluorescent dyes	Fluorescent dyes and peroxalate esters	It uses fluorescent nanoparticles to image hydrogen peroxide	High specificity and sensitivity in imaging of hydrogen peroxide	[336]
Luo et al.	Gold nanoparticles (AuNPs) and proteins are coated on PMMA and PS microtiter plates to create biocomposites that detect H_2O_2 and recombinant human interleukin-6 (rHu IL-6) using CL Enhance imaging assay. By optimizing the assay conditions, a strong linear relationship is established between CL intensity and concentrations of target analytes, making sensitive and accurate quantification easier	Gold Nanoparticles, Proteins, Poly methyl methacrylate (PMMA) plates, and PS microtiter plates	Immobilization of proteins on solid surfaces using gold nanoparticles, enhancing protein density and bioactivity	• Used in the determination of rHu IL-6 in human serum with satisfactory accuracy • Detection of H_2O_2 and rHu IL-6 with improved sensitivity and loading capacity compared to direct protein immobilization • Potential for ultrasensitive detection of various biorecognition events beyond H_2O_2 and rHu IL-6	[337]
Fan et al.	Target oligonucleotides were hybridized with magnetic bead-linked capture probes and biotin-terminated amplification DNA probes. Streptavidin-coated gold particles were then bound to the biotin probes, and the gold tags were dissolved and quantified using a luminol chemiluminescence reaction. Signal amplification was achieved by assembling biotinylated colloidal gold onto streptavidin-coated PS beads	Magnetic beads, capture probes, biotin-terminated amplifying DNA probes, streptavidin-coated gold particles, biotinylated colloidal gold, streptavidin-coated PS beads, and luminol for CL reaction	Magnetic separation/mixing process and amplification through colloidal gold labels	Ultrasensitive detection of sequence-specific DNA, and potential for sensing other biorecognition events	[338]
Som et al.	Microfluidic CL method for arsenate detection	PS-beads, vanadomolybdoarsenate heteropoly acid (VMoAs-HPA), surfactant	Sorption pre-concentration of arsenate onto PS-beads for CL detection	Rapid, sensitive detection of arsenate in various water specimens	[334]

Table 4 (continued)

Study	Methodology	Main components	Mechanism	Applications	Refs.
Kanwal et al.	A sandwich immunoassay was established using PS microspheres as a support. PS microspheres were modified with primary anti-HIgG antibody (Ab1) through electrostatic interactions. Cadmium telluride (CdTe) nanoparticles linked to horseradish peroxidase (HRP)-labeled secondary anti-HIgG antibody (Ab2) via covalent bonding were used for detection. Chemiluminescence was used for signal measurement in this assay	1. *PS microspheres*: Act as the immobilizing support and site for the immunoassay 2. *Primary anti-HIgG (Ab1)*: Captures the HIgG antigen 3. CdTe nanoparticle (CdTeNPs): Conjugated with HRP-labeled secondary anti-HIgG (Ab2) for signal amplification 4. Horseradish peroxidase (HRP): Enzyme for chemiluminescence detection	The primary anti-HIgG (Ab1) on PS microspheres captures the HIgG antigen, forming the first layer of the sandwich. CdTeNPs modified with HRP-labelled secondary anti-HIgG (Ab2) bind to the captured HIgG, creating a sandwich structure. The HRP enzyme then catalyzes a chemiluminescence reaction, amplifying the signal for detection	• High sensitivity and specificity for protein immune detection • Potential applications in DNA analysis and other biological assays • Compared to ELISA, it demonstrates good reliability, making it a feasible alternative or complementary method	[339]
Zhang et al.	CRET-based nanoparticle	Nanoparticles, luminol	Chemiluminescence Resonance Energy Transfer (CRET) for enhanced light emission	Improved detection of myeloperoxidase (MPO) in organisms	[340]
Shuhendler et al.	CL semiconducting polymer nanoparticle	Peroxyoxalate, fluorescence conjugated polymer, near-infrared dye	Combines CL and Fluorescent Resonance Energy Transfer (FRET)	Longitudinal detection of drug-induced hepatotoxicity and its remediation in mice	[341]
Acharya et al.	PSBs are functionalized with Electro-generated Chemiluminescence (ECL) labels and conjugated with monoclonal antibodies specific to the Zika virus (ZIKV), creating anti-ZIKV-PSBs. These anti-ZIKV-PSBs are used to capture ZIKV from samples, and the resulting complexes are analyzed for ECL intensity following the isolation of magnetic beads	PSBs, ECL labels, Anti-ZIKV monoclonal antibodies, and magnetic beads for isolation	An ECL-based immunoassay detects ZIKV by using antibodies allied to electro-chemiluminescent labels. When ZIKV binds to these antibodies, an electric current triggers a light-emitting response, indicating the presence of the virus with high sensitivity	- Quick and ultrasensitive detection of ZIKV in human biological fluids (saline, plasma, urine) - Potential for clinical diagnosis, vector surveillance, epidemiological, and research for labs	[342]
Yu et al.	PS/AgBr hybrid colloidal crystals for bromine gas sensing	PS, silver bromide (AgBr), silver nanoparticles (Ag-NPs)	UV-induced AgBr decomposition to Ag-NPs, enhancing structural colors	Cost-effective, recoverable colorimetric assay for the detection of bromine gas	[343]

Table 4 (continued)

Study	Methodology	Main components	Mechanism	Applications	Refs.
Chen et al.	Phtalocyanine-dyed and quantum dots (QDs)-encapsulated carboxylate-functionalizes PS nanospheres (PPs and QPs) were developed for a sandwich format immunoassay to detect carcinoembryonic antigen (CEA) in serum. Activated by a 680 nm diode laser, PPs generated singlet O_2, which interacted with QPs conjugated with antibodies via Foster Resonance Energy Transfer (FRET), emitting anti-stokes fluorescence at ~605 nm. This enabled CEA detection in 25 µL serum samples	*Phthalocyanine-dyed PS Nanospheres (PPs)*: Sensitize to produce singlet oxygen upon laser excitation *Quantum Dots-Encapsulated PS Nanospheres (QPs)*: Serve as fluorescent probes, conjugated with antibodies *680 nm Diode Laser*: Used to excite PPs *Antibodies and Antigens*: For specific detection of target biomarkers	The biosensor utilizes a CL reaction and FRET. PPs produce singlet oxygen when sensitized by the diode laser. QDs-encapsulated carboxylate-functionalized PS nanospheres (QPs), conjugated with antibodies, react with singlet O_2 and emit anti-stokes fluorescence	*Detection Limit*: Unprecedented sensitivity for carcinoembryonic antigen *Clinical Serum Detection*: Correlated well with commercial chemiluminescence immunoassay kits ($R^2 = 0.99718$, n = 107) - *Potential Uses*: Direct integration into clinical diagnostics for early detection, screening tests, and prognosis evaluation of patients	[344]
Zhou et al. 2019	Mimic-enzyme catalyst CL polymer dots (PS-hemin dots)	PS, hemin, amphiphilic polymer	Slow-diffusion-controlled reaction substantially increases CL emissions	Real-time imaging of reactive oxygen species (ROS) in bioassays and bioimaging	[333]
Liu et al. 2023	Photo-responsive hyper crosslinked polymers (HCPs) from PS waste	PS waste, azobenzene units	Photo-modulated adsorption/desorption of CO_2 for efficient carbon capture	Sustainable, low-energy carbon capture for post-combustion applications	[345]
Xing et al., 2023	The study created PEG-functionalized PS microspheres through one-step copolymerization using amphiphilic macromolecular monomers i.e. PEOVB and carboxyl monomers with styrene, incorporating photosensitizers. Characterization was done with SEM, FT-IR, and UV–visible spectroscopy. The Luminescent Oxygen Channeling Immunoassay (LOCI) performance, assessed by sensing ricin toxin (RT), revealed that microspheres with 10 wt.% PEOVB provided a higher signal-to-noise ratio and improved detection limits	1. Microspheres: PS microspheres functionalized with PEG segments (PEOVB) 2. Photosensitizer: Integrated into the microspheres 3. Protein: Bovine serum albumin (BSA) for testing non-specific adsorption [*PEOVB*: Poly (ethylene glycol) oligo (2-vinyl propyl) ether]	The author focuses on reducing non-specific protein adsorption on microspheres and increasing their performance in LOCI assays	*LOCI Detection*: Improved performance in detecting ricin toxin (RT) with enhanced signal-to-noise ratio	[346]

using heavy-oil solvents can enable PS pyrolysis at lower temperatures, around 400° Celsius, similar to catalytic pyrolysis using zeolites. PS application in the construction industry is restricted due to its deprived strain-resistant property, high flammability, and low strength. For example, incorporating PS into concrete has been shown to weaken the material. Similarly, merging HIPS and ABS in computer casings significantly reduces mechanical properties, particularly elongation to failure and impact strength. Due to the high calorific value of PS, as an energy source, its use is hampered by the release of oncogenic and noxious substances when burned at temperatures beneath 500 °C. A significant challenge in WPS involves the attitudes and observations of both industry and customers. Recycling companies often hesitate to practice post consumer resins (PCRs) or regrind resins due to the wide-ranging testing required to determine suitable applications and processing parameters [347].

The labeling of recycled resins can be misleading and lacking in detail, making it difficult to ascertain molding and mechanical properties. Assuming that castoff polymers may be degraded or mislabeled, it is important to mark the used polymer themselves rather than relying on the properties of the natural virgin polymer. One more challenge, even in advanced nations, is that current polymer databases lack information on PCRs and regrind resins. These databases are valuable for selecting materials and reducing the experimental period needed to assess molding and motorized properties. Overall, industry reluctance often arises from the financial risks and costs associated with additional testing. Besides, consumer concerns about potential public health risks from recycled plastics may also contribute to negative perceptions. Globally, the expertise in polymers is comparatively limited, a challenge that is further noticeable in the context of India due to quite a few factors. Crucial among these is the insufficient number of educational institutions offering specialized curricula in polymer sciences. Therefore, the fields of plastics, their disposal, and recycling are emerging areas of research. The abundant potential applications were discussed there is a noteworthy lack of empirical evidence regarding the widespread industrial application of these technologies in India. Yet, as per community awareness of the growing impact of plastic waste on the environment, there is an increasing expectation that reusability/recycling will be considered a possible strategy to mitigate these impacts.

11 Conclusion and future perspectives

The study highlights the transformative potential of WPS recycling with innovative approaches like the expansion of new synthetic materials like ion exchangers which offer substantial opportunities for environmental remediation. Recycling WPS and converting it into valuable products not only creates possible value chains but also eases the impact of current disposal practices on environmental degradation and public health. To maximize the benefits of recycling WPS, there is a demanding need for comprehensive policy and legal frameworks that govern the practice, disposal, and recycling of PS. This includes establishing robust systems for collection and passage, employing encouragements properly meant for disposal, and imposing taxes and penalties for inappropriate practices. Understanding user insights toward waste material products like WPS is also crucial for enhancing reutilizing efforts. Constant and current research into applications such as chemiluminescence-derived materials, lightweight concrete, ion exchangers, and composites from WPS represents promising advances.

The innovative use of PS waste in chemiluminescence applications, such as producing functionalized substrates that enhance CL reactions, showed the potential to repurpose PS into high-value, leading-edge technologies. These applications not only provide environmentally sustainable solutions but also improve the efficiency and performance of CL systems in various analytical and diagnostic fields. However, the lack of funding and facilities for research remains an important barrier, particularly in emerging nations. Promoting the reusability of PS through innovative recycling technologies and comprehensive policy measures can substantially reduce environmental impacts, contributing to the achievement of SDG 12 by promoting sustainable consumption and production patterns. Also, these efforts can support SDG 13 by mitigating climate change by reducing greenhouse gas emissions from waste.

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